hello abstract

1 Setup

$$\widehat{q}_h = a_{h,(k_h)}, \qquad k_h = \lceil (n_h+1)(1-\alpha) \rceil. \tag{1}$$

2 Envelope constructions

Each of the three envelope construction methods follows the same two stage calibration procedure: the shape-discovery set S_1 is used to determine the geometric shape of the envelope, then the size scaling set S_2 is used to determine the scaling factor $\hat{t}$. Each construction assigns a scalar score, $\tau(v,m)$, to a score vector v with observed set m. A candidate class is then included in the prediction set when $\tau(v,m) \le \hat{t}$, where $\hat{t}$ is calculated on S_2 as in Section 1. For both the collapsed and radial methods, we define $\tau(v) = \inf\{t > 0 : v \in t \cdot E\}$ using an envelope E which is star shaped with respect to the origin and give a closed form expression for it. The strip method works in the same way, except that the single scaling factor $\hat{t}$ multiplies the limits of the staircase rather than dilating the whole region, so τ is defined directly as a maximum of ratios.

We first fix a class h and drop the index h. K denotes the number of coordinates that remain after removing the coordinates that are missing for every sample of S_1. For a vector $v \in [0,\infty)^K$ and observed set $m \subseteq \{1,\dots,K\}$, we write v_k, $k \in m$, for just its observed entries. We use $\varepsilon = 10^{-12}$ to prevent division by zero, and $Q_p(\cdot)$ denotes the empirical p-quantile.

Collapsed envelope. The basic envelope construction simply collapses the K-dimensional score vector to a scalar score by taking the mean over the observed coordinates, and calibrates a single quantile on S_1:

$$\bar{v} = \frac{1}{|m|}\sum_{j\in m} v_j, \qquad \tilde{q} = Q_{1-\alpha}\big(\{\bar{v}_i\}_{i\in S_1}\big), \qquad \tau(v,m) = \frac{\bar{v}}{\tilde{q}}, \qquad E = \{v : \bar{v} \le \tilde{q}\}. \tag{2}$$

This method discards the geometric information that the K coordinates provide. Two vectors with the same mean are treated identically whatever the distribution of their coordinates is, so a single coordinate with strong evidence (a very small v_k) is averaged together with all the others. The accepted region is a halfspace in the observed coordinates. It serves as a mean score baseline for the multidimensional constructions below. After calibration on S_2, a vector is accepted when $\bar{v}/\tilde{q} \le \hat{t}$, which is the same as $\bar{v} \le \hat{t}\,\tilde{q}$. Here $\hat{t}$ is the k-th smallest value of $\bar{v}_i/\tilde{q}$ over S_2, so $\hat{t}\,\tilde{q}$ is just the k-th smallest value of $\bar{v}_i$ over S_2.

Radial envelope. Let $u_1,\dots,u_M$ be unit vectors in the nonnegative orthant ($M = 250$ by default), obtained by normalizing the absolute value of Gaussian vectors with a fixed random seed. For $s \in S_1$ with observed set m_s, let $\|s\|$ be the Euclidean norm of its observed entries and $\hat{s}$ its direction (zero on missing entries). For each r, let $N_r = \{s \in S_1 : \langle \hat{s}, u_r\rangle \ge \cos\theta\}$ with $\theta = 30^\circ$, and set

$$\rho_r = Q_{1-\alpha}\big(\{\|s\| : s \in N_r\}\big) \tag{3}$$

when $|N_r| \ge n_{\min} = \max\{10, \lfloor 0.2\,|S_1|\rfloor\}$. If $0 < |N_r| < n_{\min}$, ρ_r is the convex combination of the local quantile and the marginal quantile $Q_{1-\alpha}(\{\|s\| : s \in S_1\})$ with weight $|N_r|/n_{\min}$ on the local one, and if $N_r = \emptyset$ it is the marginal quantile. A smoothed radius in direction a is

$$R(a) = \sum_{r=1}^{M} w_r(a)\,\rho_r, \qquad w_r(a) = \frac{\exp(\kappa\langle a, u_r\rangle)}{\sum_{r'}\exp(\kappa\langle a, u_{r'}\rangle)}, \tag{4}$$

with $\kappa = 8$. Evaluating it at the anchors $a_t \in \{u_1, \dots, u_M, e_1, \dots, e_d\}$ (the e_j are the coordinate axes) gives boundary points $b_t = R(a_t)\, a_t$, and the score is

$$A^{\mathrm{rad}}(v, m) = \min_t \max_{j \in m} \frac{v_j}{b_{tj} + \varepsilon}. \tag{5}$$

Since $A^{\mathrm{rad}}(v, m) \le q$ means that $v_j \le q\, b_{tj}$ for all observed j for some t, the accepted region is a union of boxes with a corner at $q\, b_t$, restricted to the observed coordinates.

Strip envelope. For every coordinate j, the interval from the smallest to the largest observed S_1 value is cut into $B = 8$ bins of equal width, and $b_j(x) \in \{1, \dots, B\}$ is the bin of x (values outside the interval go to the end bins). Let $\alpha' = \alpha / d$. For each ordered pair $j \ne i$ and each bin b, the limit

$$L_{ji,b} = Q_{1-\alpha'}\big(\{s_i : s \in S_1,\ s_j, s_i \text{ observed},\ b_j(s_j) = b\}\big) \tag{6}$$

is the quantile of coordinate i among S_1 samples whose coordinate j falls in bin b. Bins with fewer than 3 samples are blended with the marginal quantile of coordinate i in the same way as in the radial case. The limits are then made nonincreasing in the bin index,

$$\tilde{L}_{ji,b} = \max_{b' \ge b} L_{ji,b'}, \tag{7}$$

and the score is

$$A^{\mathrm{strip}}(v, m) = \begin{cases} \max\limits_{\substack{j, i \in m \\ j \ne i}} \dfrac{v_i}{\tilde{L}_{ji, b_j(v_j)} + \varepsilon}, & |m| \ge 2, \\ \dfrac{v_j}{q_j + \varepsilon}, & m = \{j\}, \end{cases} \tag{8}$$

where $q_j = Q_{1-\alpha'}(\{s_j : s \in S_1 \text{ observed}\})$. The level α/d is a Bonferroni-type choice for the d coordinates. It affects the shape of the score only: the coverage of the final set is set by $\widehat{q}$. Because the bin is chosen from the unscaled value v_j, the thresholded regions are score sublevel sets and not dilations of one fixed region.

Properties. Table 1 lists the main differences between the methods. The three scores share one structural property.

Proposition 2.1. *Fix an observed set m and let v, v' have the same observed set with $v'_j \le v_j$ for all $j \in m$. Then $A(v', m) \le A(v, m)$ for each of the three scores. Hence every region $\{A \le \widehat{q}\}$ is downward closed: if a label is accepted, it stays accepted when its scores improve.*

Proof. For (2) this is clear because the mean is nondecreasing in each coordinate. In (5) each ratio $v_j/(b_{tj} + \varepsilon)$ is nondecreasing in v_j, and a maximum and a minimum of nondecreasing functions is nondecreasing. For (8), $v'_j \le v_j$ gives $b_j(v'_j) \le b_j(v_j)$, so $\tilde{L}_{ji, b_j(v'_j)} \ge \tilde{L}_{ji, b_j(v_j)}$ by (7). Together with $v'_i \le v_i$ every ratio in (8) gets smaller. The case $|m| = 1$ is again a ratio nondecreasing in v_j. □

The proposition is not used for the coverage statement. For fixed S_1 (and fixed directions u_r, which are drawn independently of S_2 and of the new observation) each score is a fixed function of (v, m), so the rank argument of Section 1 applies to all three in the same way. Monotonicity is only a sanity property of the learned shapes.

	Collapsed	Radial	Strip
Learned on S_1	kept coordinates	radii ρ_r	limits $\tilde{L}_{ji,b}$, quantiles q_j
Region (fixed m)	halfspace	union of boxes	intersection of pairwise constraints
Downward closed	yes	yes	yes
Single observed coordinate	the value itself	ratio to the boundary points	ratio to the marginal quantile q_j
Fully missing vector at prediction	label rejected	label accepted	label accepted

Table 1: Comparison of the three constructions. A sample with no observed score for the true class is rejected during fitting, so the rule for fully missing vectors at prediction is not covered by the calibration, and we make no coverage claim for it.